Day 12 Activity: A `flights` investigation

Wednesday, June 3, 2026

Your Name Here

Setup

```
library(tidyverse)
library(nycflights13)
```

Today we use `flights`, every flight out of NYC in 2013. Glimpse it, and note that cancelled flights are denoted using `NA` missing values in the `dep_delay` and `arr_delay` columns.

```
glimpse(flights)
```

```
Rows: 336,776
Columns: 19
$ year           <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
$ month          <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ day            <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ dep_time       <int> 517, 533, 542, 544, 554, 554, 555, 557, 557, 558, 558, ~
$ sched_dep_time <int> 515, 529, 540, 545, 600, 558, 600, 600, 600, 600, 600, ~
$ dep_delay      <dbl> 2, 4, 2, -1, -6, -4, -5, -3, -3, -2, -2, -2, -2, -2, -
1~
$ arr_time       <int> 830, 850, 923, 1004, 812, 740, 913, 709, 838, 753, 849,~
$ sched_arr_time <int> 819, 830, 850, 1022, 837, 728, 854, 723, 846, 745, 851,~
$ arr_delay      <dbl> 11, 20, 33, -18, -25, 12, 19, -14, -8, 8, -2, -3, 7, -
1~
$ carrier        <chr> "UA", "UA", "AA", "B6", "DL", "UA", "B6", "EV", "B6", "~
$ flight         <int> 1545, 1714, 1141, 725, 461, 1696, 507, 5708, 79, 301, 4~
$ tailnum        <chr> "N14228", "N24211", "N619AA", "N804JB", "N668DN", "N394~
$ origin         <chr> "EWR", "LGA", "JFK", "JFK", "LGA", "EWR", "EWR", "LGA",~
$ dest           <chr> "IAH", "IAH", "MIA", "BQN", "ATL", "ORD", "FLL", "IAD",~
$ air_time       <dbl> 227, 227, 160, 183, 116, 150, 158, 53, 140, 138, 149, 1~
```

```
$ distance       <dbl> 1400, 1416, 1089, 1576, 762, 719, 1065, 229, 944, 733, ~
$ hour           <dbl> 5, 5, 5, 5, 6, 5, 6, 6, 6, 6, 6, 6, 6, 6, 6, 5, 6, 6, 6~
$ minute         <dbl> 15, 29, 40, 45, 0, 58, 0, 0, 0, 0, 0, 0, 0, 0, 0, 59, 0~
$ time_hour      <dttm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 2013-01-
01 0~
```

Today, we will practice using the six verbs (`filter`, `arrange`, `select`, `mutate`, `group_by`, `summarise`), plus `count()` and the pipe to chain them together.

Part 1 — Plan before you code

For this question: *Which of the three NYC airports (`origin`) sent out the most flights in 2013?*

1a. First, write out your plan in English using one sentence per step. Answer the following questions:

1. Which verbs will you need to use, and in what order?
2. What does one row of your target table represent?

   [Write your plan here.]

1b. Now implement your plan with code. (Recall: `count()` is a shortcut for a `group_by()` + `summarise()`. Both give the same result.)

```
# Your code:
```

Part 2 — Build a pipeline

Which carrier had the worst average arrival delay, among carriers with at least 1,000 flights?

Build a pipeline step by step; filter is **not** needed on the raw rows here, but you'll need it on the **summary table** to drop small carriers. Include `n = n()`.

```
# Your code:
```

Which carrier is it, and how many flights is that average based on?

[Write your answer here. Note that we only have the 2-letter code for carrier in today's data; turning it into the airline's name needs a *join*, which we've seen but haven't covered in detail yet.]

Part 3 — mutate(), then plot

3a. Create a new variable `time_of_day` that takes on the value `"morning"` when `dep_time < 1200` and `"afternoon/evening"` otherwise. Then find the average departure delay for each, with a count.

```
# Your code:
```

3b. Pipe a version of that summary into a `geom_col()` plot of average delay by `time_of_day`.

```
# Your code:
```

In one sentence, explain what your plot shows:

[Write your answer here.]

Part 4 — Don't get fooled

4a. Order matters. These two pipelines run without error but answer **different questions**. Run both, then explain in one or two sentences how the questions differ.

```
# Pipeline A
flights |>
  filter(dest == "BOS") |>
  summarise(avg = mean(arr_delay, na.rm = TRUE))
```

```
# Pipeline B
flights |>
  group_by(dest) |>
  summarise(avg = mean(arr_delay, na.rm = TRUE)) |>
  filter(dest == "BOS")
```

[Write your answer here. Address the following: how do the questions differ? Do the numbers match? Should they?]

4b. Spot the silent `NA`. This returns `NA` for some carriers. Why? What single change fixes it?

```
flights |>
  group_by(carrier) |>
  summarise(avg_delay = mean(dep_delay))
```

[Write your explanation and the fix in words here.]

Part 5 — Open investigation

Pick **one** of these questions (or invent your own), and answer it with a pipeline:

- Which destination has the most *variable* arrival delays? (Hint: `sd()` is a reduction, like `mean()`.)
- Do flights later in the day make up time in the air more than morning flights?
- Which day of the week (you would needthe `month` and `day` variables) is worst to fly? *(Stick to month if day of the week is too much.)*

5a. State your question and your English plan.

[Write your question & plan here.]

5b. Your pipeline:

```
# Your code:
```

5c. Report your checks. Answer the following three questions:

1. How did you handle missing values?
2. What's the smallest sample size `n` in your result, and is it big enough to trust?
3. Does the answer match your intuition (what you would expect)?

    1. [Write your answer here.]
    2. [Write your answer here.]
    3. [Write your answer here.]

Code appendix

```r
library(tidyverse)
library(nycflights13)
glimpse(flights)
# Your code:

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# Your code:

# Pipeline A
flights |>
  filter(dest == "BOS") |>
  summarise(avg = mean(arr_delay, na.rm = TRUE))
# Pipeline B
flights |>
  group_by(dest) |>
  summarise(avg = mean(arr_delay, na.rm = TRUE)) |>
  filter(dest == "BOS")
flights |>
  group_by(carrier) |>
  summarise(avg_delay = mean(dep_delay))
# Your code:
```